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## **Identification of Polymer Flooding Flow Channels and Characterization of Oil Recovery Factor Based On U-Net**

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### **Abstract**

Image identification is a major means to achieve quantitative characterization of the microscopic oil displacement process. Traditional digital image processing techniques usually use a series of pixel-based algorithms, which is difficult to achieve real-time processing of large-scale images. Deep learning methods have the characteristics of fast speed and high accuracy. This paper proposes a four-channel image segmentation method based on RGB color and rock particle mask. First, the micro model rock particle mask is divided together with the RGB component to form four-channel input data through image processing technology. Pixel-level training set labels are then created through traditional image processing techniques. Through the U-Net semantic segmentation network, the pixel-level oil and water identification and recovery factor calculation of the polymer microscopic oil displacement process were carried out. Combined with the pore distance transformation algorithm, the lower limit of pore utilization for different displacement media was clarified. The results show that U-Net can achieve accurate division of oil and water areas. Compared with conventional three-channel images, the improved four-channel image proposed in this paper has significantly improved the segmentation accuracy due to the addition of the constraints of the rock particle mask, and the global accuracy can be up to 99%. Combining some post-processing methods, this paper found that polymer flooding increased the mobilization degree of small pores on the basis of water flooding and lowered the lower limit of pore mobilization from 25  $\mu\text{m}$  to 16  $\mu\text{m}$ . In microscopic experiments, the recovery factor was increased by 24.01%, finally achieving rapid and accurate quantitative characterization of the microscopic oil displacement process. The four-channel image method based on the U-Net semantic segmentation network and the improved rock particle mask proposed in this article has strong adaptability to the identification of flow channels in the microscopic oil displacement process. Quantitative characterization of the lower limit of pore movement and recovery degree during microscopic oil displacement provides a new method for microscopic image processing.

## Introduction

Remaining oil in reservoirs can be divided into macroscopic and microscopic categories. Macroscopic remaining oil is typically in regions that are difficult to sweep due to geological structures or engineering reasons, while microscopic remaining oil is due to factors such as pore properties within the swept region (Wang et al., 2021). The characteristics of macroscopic remaining oil are the overall performance of microscopic remaining oil (Weichao & Sun, 2016). The research methods for microscopic remaining oil mainly include two categories: experimental methods (Gao et al., 2022) and numerical simulations (Yang et al., 2021). Experimental methods include traditional optical techniques (fluorescent sheet (Wei et al., 2023), true core models, and microfluidic technology (Chen et al., 2023)), Computed Tomography CT technology (Wang et al., 1984), Nuclear Magnetic Resonance (NMR) technology (Wei et al., 2020), Scanning Electron Microscopy (SEM) technology (Farhadi et al., 2022), and Confocal Laser Scanning Microscopy (CLSM) (Li et al., 2014). Microfluidics is a technology that involves the use of micro-sized channels to process or manipulate tiny volumes of fluids. It is an emerging interdisciplinary field that crosses over chemistry, fluid physics, microelectronics, and new materials. With the help of microscopes and cameras, microfluidics offers convenient, repeatable, and real-time monitoring capabilities (Yu et al., 2019). It has found increasingly widespread applications in the oil industry (Cao et al., 2023; Ghosh & Schurtenberger, 2019). Current reports on the application of microfluidics in the oil industry often provide qualitative or simple overall analyses of images (Liu et al., 2022; Ma et al., 2022; Yekeen et al., 2023), lacking a quantitative and detailed description of remaining oil distribution characteristics from the image perspective.

With the continuous development of deep learning image segmentation techniques (Garcia-Garcia et al., 2018), images obtained from microfluidic experiments can also be segmented using deep learning algorithms, providing a basis for subsequent quantitative analysis. Pixel-based semantic segmentation of microfluidic images is used to determine the location distribution of oil, water (polymers), and rock particles in a microscopic model at a given moment. Researchers have proposed many pixel-based semantic segmentation networks, such as Fully Convolutional Networks, U-Net, DeepLab, RefineNet, SegNet, and variations of these models (Badrinarayanan et al., 2017; Chen et al., 2017; Lin et al., 2017; Ronneberger et al., 2015), among which U-Net is widely applied in medical image segmentation and has also been used in pore identification in scanning electron microscopy images (Bangaru et al., 2022), fracture extraction in CT images of shale (X. Wu et al., 2023), and mineral identification in core images (Li et al., 2022).

This article proposes a four-channel U-Net with a rock particle mask to accurately distinguish oil, water (polymer), and rock particles in microfluidic images. Combined with image processing techniques, it achieves quantitative characterization of oil-water distribution at pore scales, providing new ideas for the processing of microscopic images.

## Method

### Image preprocessing and data labeling

Compared to conventional computer vision datasets, which require extensive preprocessing to eliminate the influence of external factors, microfluidic experiments often pay attention to these factors during filming, resulting in high-quality raw images with little noise that can be directly processed. The initial saturated oil state microscopic model only includes oil (pore area) and rock area. In this research, the Otsu (Otsu, 1979) method based on maximum between-class variance is used to extract the threshold for segmentation of the initial saturated oil grayscale image. As shown in the Fig. 1, the black area is the pore area filled with oil, and the white part is the rock area. The code used in this research is based on Matlab software.

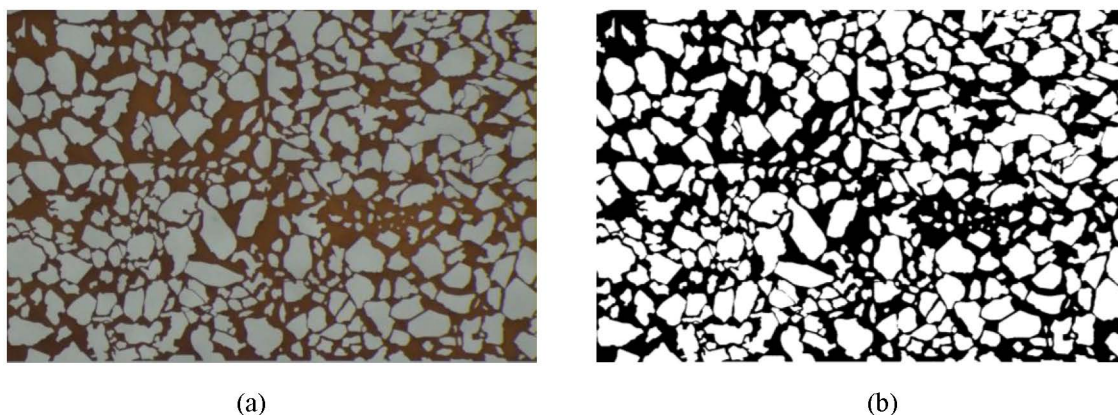


Figure 1—(a) Original saturated oil image and (b) rock-pore region after binarization

After obtaining the binarized image, a series of morphological processing (Eddins, 2014) is required to improve the quality of the image. First, areas with less than 15 pixels in the image are filled in, then the image is subjected to opening operation to eliminate small objects, and the boundary adhesion area of rock particles at the thin points is separated. The boundary of the larger rock particle area is smoothed without significantly changing the area ratio of the rock/pore area.

In order to obtain a train label, images from the late stage of polymer flooding are selected for training. For microfluidic images, only the black and white information of the image needs to be retained. The difference between the binary images of the initial and final states can be directly calculated, but this result is often unstable. Therefore, this research decides to directly calculate the difference between the grayscale images of the two images (Fig. 2a). The image is binarized and the area (Fig. 2b) with less than 100 pixels in the binary image is filtered out.

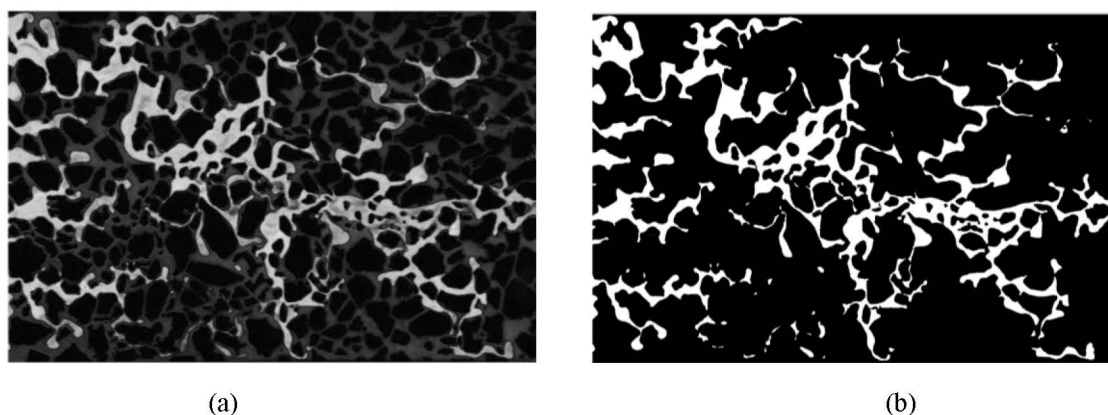


Figure 2—(a) the grayscale difference map; (b) the binary map (right) at the initial-current time

Combining the initial oil-water distribution and modifying the swept area, the desired training labels can be obtained. As shown in Fig. 3, before determining the final training labels, some uncertain areas need to be manually corrected based on the image.



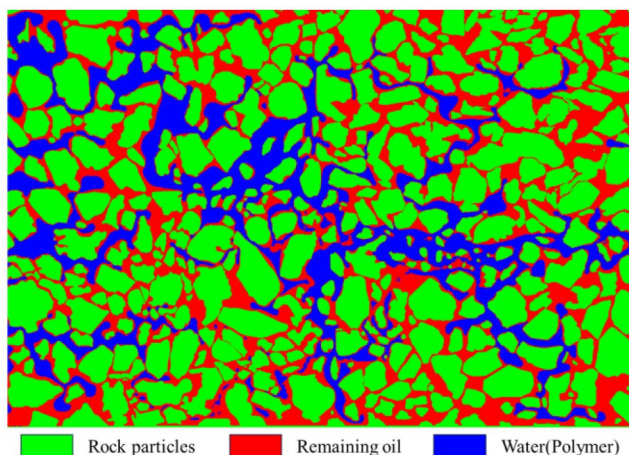


Figure 3—U-Net training label

### Image segmentation method based on deep learning

U-Net is a convolutional neural network for image segmentation, consisting of an encoder and a decoder (Ronneberger et al., 2015), as shown in Fig. 4. The encoder part is composed of multiple convolutional layers and downsampling layers, which extract features from the input image. The decoder part is used to restore the spatial information of the original image from the feature maps obtained by the encoder. The decoder part includes multiple upsampling layers and convolutional layers (Bangaru et al., 2022). In the U-Net network structure, skip connection is also used (Chudasama et al., 2024). The feature maps from the encoder are directly connected to the corresponding positions in the decoder to provide more contextual information, allowing the network to better learn contextual information during training and improve segmentation accuracy.

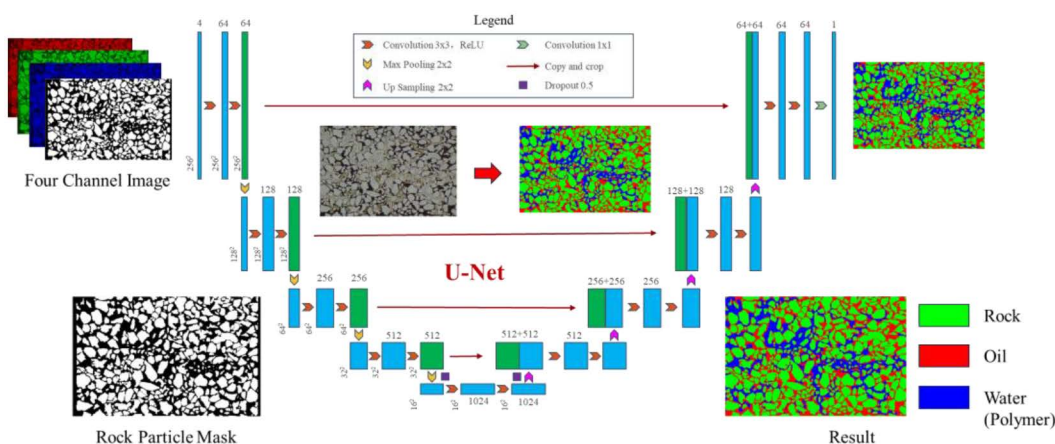


Figure 4—Flow chart of U-Net segmentation four channel image (The four channel image input in the figure includes the RGB three channels of the original image and the rock particle mask channel. The U-Net semantic segmentation network is trained through labeled image, and the segmented image includes three region labels: rock, residual oil, and polymer)

Based on obtaining training labels, U-Net training is carried out. First, a four-channel image is created by adding a rock particle mask to the RGB three-channel image, which is used as input data for the network. Taking into account computer performance and computational time costs, a small batch deep learning approach is adopted.

In this research, when training the model, the training set is divided into 256x256 pixel images for training, and data augmentation is applied using techniques such as translation and rotation to expand the training set to 16000 images. The minibatch size is set to 16, and training is performed for 150 epochs. The

initial learning rate is set to 0.05, and every 10 epochs, the learning rate is reduced by a tenth. In addition, we shuffle the training data before each training epoch. We use SGD with momentum (SGDM) to train the 4-channel U-Net established in this research.

### Evaluation methods for training effectiveness

**Class accuracy (CA).** Class accuracy represents the percentage of pixels correctly classified for each class. For each class, the ratio of correctly classified pixels to the total number of pixels in that class is the class accuracy.

$$\text{Class accuracy} = \frac{TP}{TP + FN} \times 100\% \quad \text{Eq.1}$$

Where TP is true positive, and FN is false negative.

By averaging the CA of different classes, we obtain the mean class accuracy (MCA), which describes the overall accuracy of the image.

**Boundary F1 (BF) score (BS).** The BF score is an indicator that measures the alignment of predicted class boundaries with true boundaries. It is defined as the harmonic mean of precision and recall, and has a tolerance for distance errors to determine whether points on the predicted boundary match the true boundary (Csurka et al., 2013).

$$\text{BF score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{recall} + \text{precision}} \quad \text{Eq.2}$$

Similarly, we can obtain the mean BF score (MBS) for images.

**Global accuracy (GA).** Global accuracy represents the proportion of correctly predicted pixels out of the total number of pixels in the image.

**Intersection over union (IoU).** IoU score is a standard performance metric for object class segmentation problems, which can be simply understood as overlap, as shown in Fig. 5. Given a set of images, IoU measures the similarity between the predicted region and the actual region of objects present in the images (Rahman & Wang, 2016). Similarly, we obtain the mean intersection over union (MIOU).

$$\text{IoU} = \frac{TP}{FP + TP + FN} \times 100\% \quad \text{Eq.3}$$

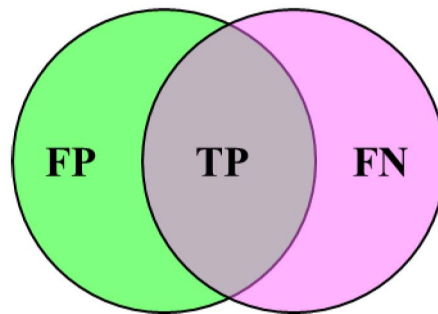


Figure 5—Intersection over union (IoU) principle diagram (where TP is true positive, FN is false negative is false negative)

### Evaluation methods for recovery efficiency

When identifying the overall recovery efficiency of the model, it is only necessary to count the difference in pixels occupied by the oil phase between the initial and end times of displacement. This is calculated by subtracting the number of pixels occupied by the oil phase at the end of displacement from the number of pixels occupied by the oil phase at the beginning, and then dividing the result by the number of pixels occupied by the oil phase at the beginning.

$$RF = \frac{NP_{oil}^{ini} - NP_{oil}^{end}}{NP_{oil}^{ini}} \quad \text{Eq.4}$$

Where RF is recovery factor,  $NP_{oil}^{ini}$  represents the number of pixels occupied by the oil phase at the beginning, and  $NP_{oil}^{end}$  represents the number of pixels occupied by the oil phase at the end of displacement.

The overall idea for evaluating pore-level sweep efficiency is to first divide the image area corresponding to pores of different radius, and then obtain the sweep efficiency of pores of different radius based on the oil phase pixel ratio in the initial and end times of displacement in each region. The overall technical route is shown in Fig. 6:

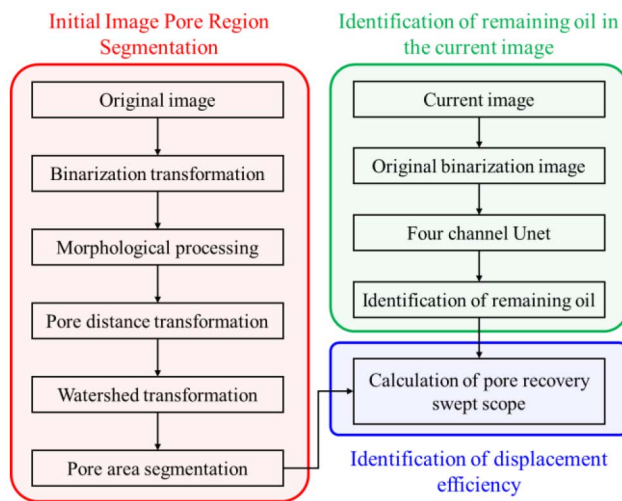


Figure 6—Roadmap for evaluating the sweep efficiency of different radius pores (The part within the red box indicates that a series of processing algorithms are applied to the image at the initial time to obtain image regions of different radius pores. The part within the green box indicates that the remaining oil region at the current time is divided using the 4-channel U-Net algorithm proposed previously. The part within the blue box indicates that the sweep efficiency of different radius pores is evaluated using image regions of different radius pores, as well as the initial and current oil regions)

In this article, a segmentation method based on pore distance transformation and watershed algorithm is used to analyze the image areas corresponding to pores of different radius. This segmentation method is applied to the initial image. As mentioned earlier, relevant morphological processing needs to be carried out. First, pore distance transformation is carried out to obtain a distance distribution map of each pixel in the pore region to the nearest rock particle. This research refers to the distance representation method (Eq. 5) proposed by Maurer (Maurer et al., 2003) and related algorithms and completes pore distance transformation with Matlab software.

$$A(x, y) = \left( \sum_{i=1}^k w_i (x_i - y_i)^p \right)^{\frac{1}{p}} \quad \text{Eq.5}$$

Where  $x$  and  $y$  are  $k$ -tuples,  $x_i$  and  $y_i$  are the  $i$ th coordinates of  $x$  and  $y$ ,  $w_i$  is the weight of the  $i$ th coordinates of  $x$  and  $y$ , and  $1 \leq p \leq \infty$ . Considering the actual physical meaning, the value of  $p$  in the formula should be 2, which means the Euclidean distance is applied.

The final result is shown in Fig. 7, where the white area represents rock particles and does not participate in the calculation. The maximum pore radius in this experimental model is about 185.98  $\mu\text{m}$ . It should be noted that the model used in this research is 6000  $\mu\text{m}$  long and 4000  $\mu\text{m}$  wide. When shooting, the pixels in both directions are 3239 and 2150, respectively. Therefore, the average pixel distance of 1.8564  $\mu\text{m}/\text{pixel}$  in both directions is used as the scale factor in subsequent algorithms.



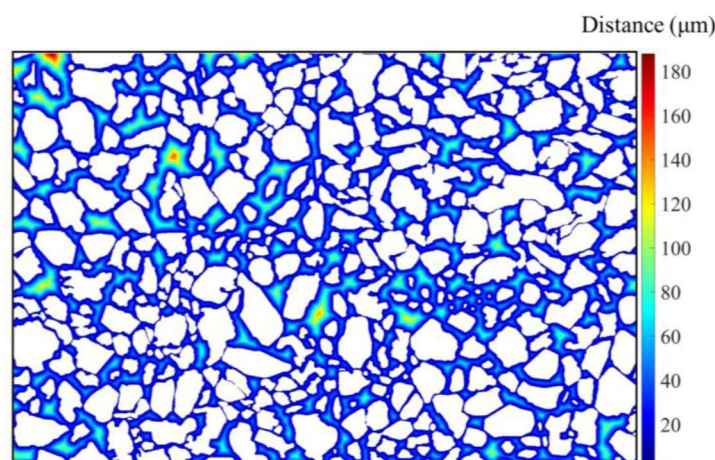


Figure 7—Pore distance transformation results (white indicates rock particles)

Then, the watershed algorithm is used to extract the axis position of pores (Fig. 8a, Fig. 8b), which determines the minimum value position in the pore distance transformation result image through the watershed algorithm (Meyer, 1994). After determining the axis line position, a specific range of axis radius is selected, and pore radius dilation (Fig. 8c, Fig. 8d) is carried out (Gao et al., 2021) to determine the control area of pores of different radius.

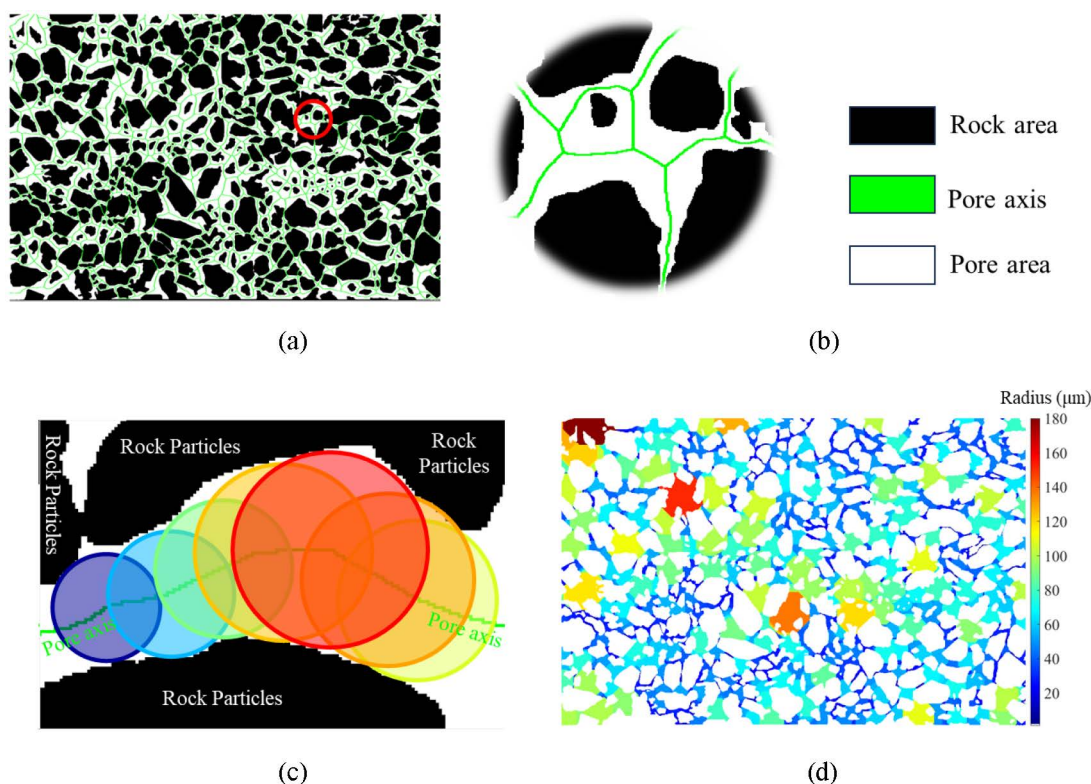


Figure 8—Segmentation method of different radius pore control areas ((a) overall view of the pore axis; (b) enlarged view of the pore axis; (c) schematic diagram of the principle of pore radius dilation; (d) segmentation result diagram of different radius pore control areas)

Finally, for a specific radius range of pore control area, the goal of evaluating pore sweep efficiency can be achieved by comparing the change in oil-occupied pixel number before and after displacement.

$$RF_i = \frac{NP_{oil}^{ini} - NP_{oil}^{end}}{NP_{oil}^{ini}} \quad \text{Eq.6}$$

Where  $RF_i$  is the recovery factor for pores in the  $i$ -th radius range,  $NP_{oil}^{ini}$  represents the number of pixels occupied by oil in that radius range at the beginning, and  $NP_{oil}^{end}$  represents the number of pixels occupied by oil in that radius range at the end of displacement.

## Result

### Comparison of the results of four-channel U-Net and three-channel U-Net

From the global accuracy and cross entropy loss during the training process (Fig. 9), it can be seen that the training speed of the 4-channel U-Net is faster than that of the 3-channel U-Net. The model performance of the 4-channel U-Net is basically stable after 500 iterations, while the traditional model requires more than 1500 iterations, and the final accuracy of the 4-channel U-Net is higher.

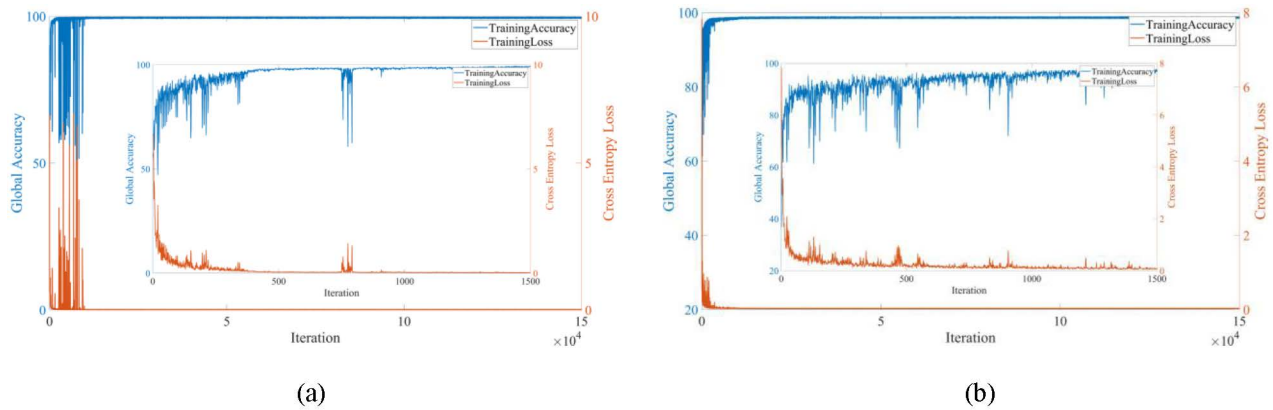


Figure 9—Comparison of model training results: (a) 4-channel U-Net training curve; (b) 3-channel U-Net training curve;

Through the pictures of displacement process, it can be found that the segmentation effect of 4-channel U-Net is better than that of 3-channel U-Net. As shown in Fig. 10a-10c, the overall segmentation results of the two methods are relatively good, and the distribution areas of oil, water (polymer), and rock particles can be basically accurately segmented. The overall segmentation accuracy rate is above 90%. According to the confusion matrix of the segmentation results of the two methods, it can be found that for each class, the recognition accuracy of 3-channel U-Net is above 90%, and a considerable part of oil and water (polymer) is identified as rock particles. The IoU of the rock class is 0.9624, resulting in relatively low segmentation accuracy of the real oil class and polymer class (the 2, 4, 5 parts in Fig. 10f-10h). Similarly, there are also some rocks that are truly identified as other classes, and the IoU of the water (polymer) class is only 0.8941. The recognition accuracy of 4-channel U-Net is basically above 99.5%, and the introduction of rock particle masks significantly improves the recognition accuracy. The probability of misidentifying oil and water (polymer) as rock particles has been significantly reduced. The overall IoU of the three classes is above 0.98, and the probability of rock particles predicted as true class is even 100%. In addition, it can be found from the comparison figure that there is pixel confusion in the segmentation results of 3-channel U-Net (the 1, 2, 3 parts in Fig. 10f-10h), and there will be abnormal segmentation boundaries (the 4 part in Fig. 10f-10h). Overall, the segmentation effect of 4-channel U-Net is better, and it has stronger adaptability for microfluidic images. The specific model performance is shown in Table 1 and Table 2.



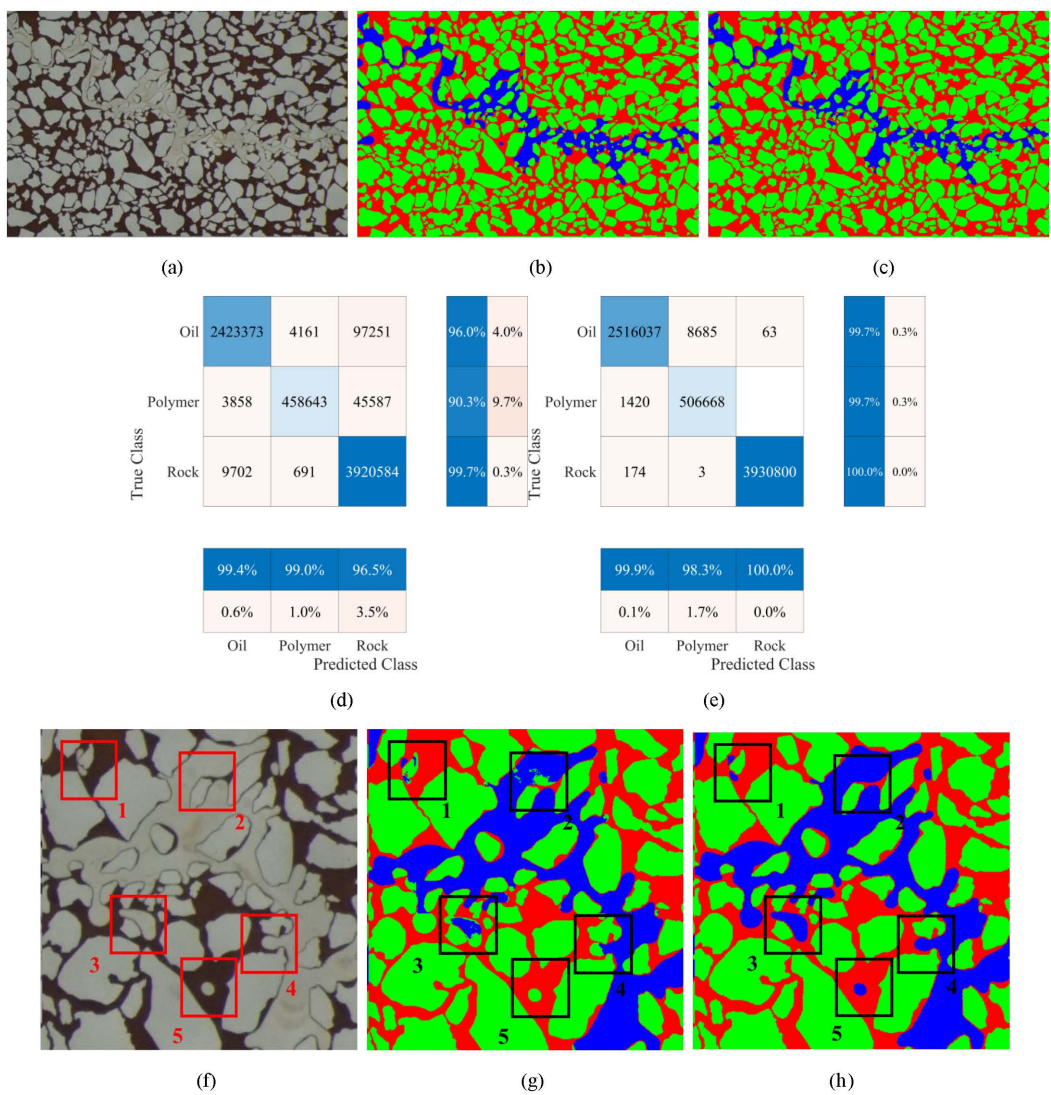


Figure 10—Comparison of the original image and the segmentation results of different methods ((a) represents the original image; (b) represents the segmentation result of 4-channel U-Net; (c) represents the segmentation result of 3-channel U-Net; (d) represents the confusion matrix of the segmentation result of 4-channel U-Net; (e) represents the confusion matrix of the segmentation result of 3-channel U-Net; (f) represents the enlarged local image of the original image; (g) represents the enlarged local image of the segmentation result of 3-channel U-Net; (h) represents the enlarged local image of the segmentation result of 4-channel U-Net)

Table 1—Comparison of overall segmentation effects of different methods

|                 | MCA    | MBS    | GA     | MIoU   |
|-----------------|--------|--------|--------|--------|
| 4-channel U-Net | 0.9979 | 0.9991 | 0.9985 | 0.9921 |
| 3-channel U-Net | 0.9533 | 0.9894 | 0.9768 | 0.9371 |

Table 2—Comparison of class segmentation effects of different methods

|     | 4-channel U-Net |        |         | 3-channel U-Net |        |         |
|-----|-----------------|--------|---------|-----------------|--------|---------|
|     | Oil             | Rock   | Polymer | Oil             | Rock   | Polymer |
| CA  | 0.9965          | 1      | 0.9972  | 0.9598          | 0.9974 | 0.9027  |
| BS  | 0.9999          | 1      | 0.9975  | 1               | 0.9950 | 0.9732  |
| IoU | 0.9959          | 0.9999 | 0.9804  | 0.9547          | 0.9624 | 0.8941  |

### Evaluation of recovery factor and pore sweep efficiency

Based on the 4-channel U-Net model described above, a set of images from a microscopic displacement process were analyzed (Fig. 11). The microscopic displacement experiment first performed water flooding, followed by polymer flooding. The results showed that polymer flooding could increase the recovery factor by 24.01% on the basis of water flooding. The process recovery factor results during displacement are shown in Fig. 12.

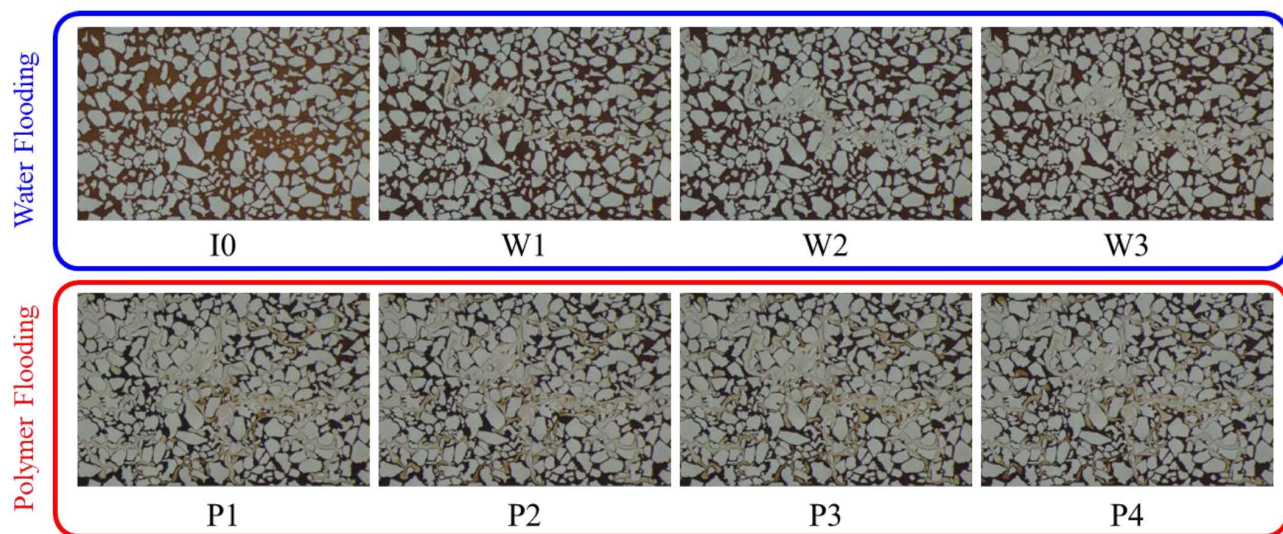


Figure 11—Microfluidic experiment whole process image

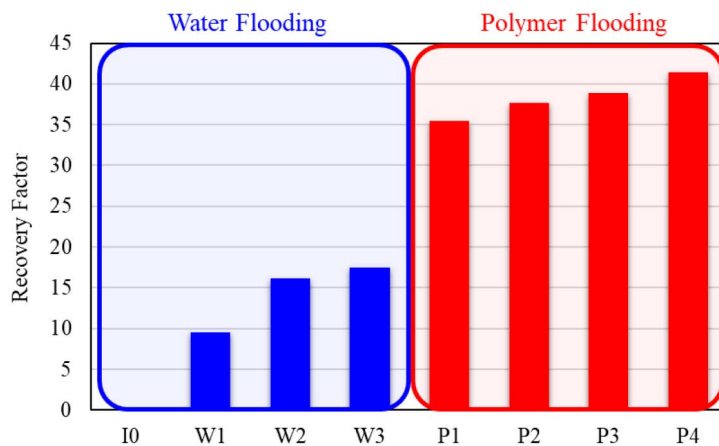


Figure 12—Microfluidic experimental recovery factor histogram based on 4-channel U-Net

Analyzing the sweep efficiency of different radius of pores during displacement process, it can be found that large and medium pores are mainly swept during water flooding process, and the cross flow is serious during water flooding process. With the increase of displacement time, the sweep efficiency of small pores increases, and the lower limit of swept pore decreases. By the end of water flooding, the lower limit of swept pore is about 25 $\mu$ m. Polymer flooding significantly improves the sweep efficiency of all levels of pores and throats, improves the uniform sweep efficiency of displacement phase, and makes the sweep efficiency more uniform. At the same time, the lower limit of swept pore further decreases, and by the end of polymer flooding, the lower limit of swept pore is about 16 $\mu$ m.

This study uses a pore area mask as the fourth channel for UNet input data, but the segmentation method of pore masks is still limited by traditional image processing techniques, but overall, the accuracy meets

the requirements. The next main research direction is how to obtain high-quality pore area mask and their application in complex chemical flooding.

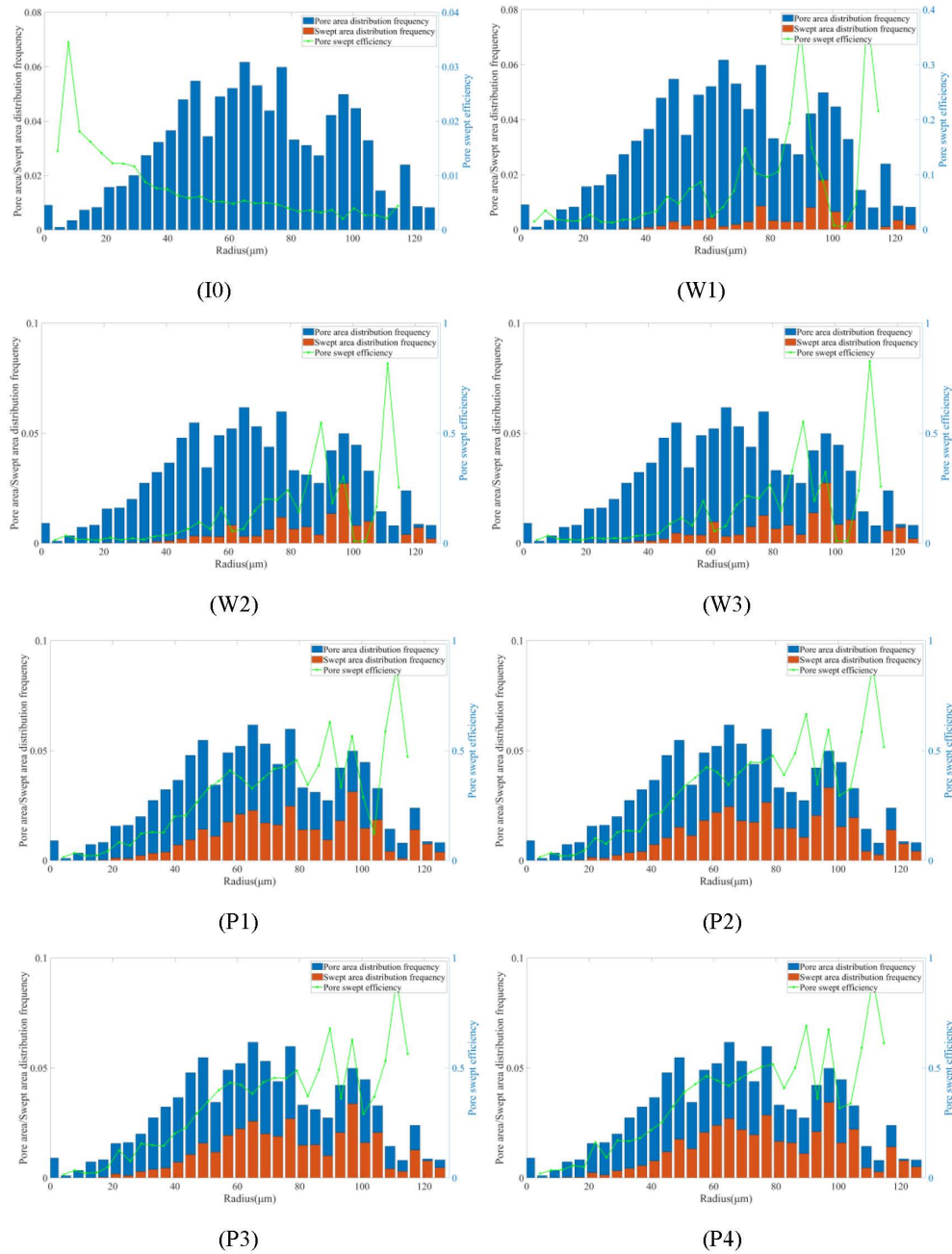


Figure 13—Distribution diagram of pore area/swept area and pore sweep efficiency with different radius((I0) represents the initial time; (W1-W3) represents the pore sweep efficiency during water flooding; (P1-P4) represents the pore sweep efficiency during polymer flooding)

## Conclusion

1. This study proposes a microscopic image segmentation method based on a modified 4-channel U-Net with a rock particle mask. Compared to traditional methods, the processing speed has been significantly improved. The constraint of the rock particle mask has resulted in a higher accuracy compared to traditional methods, significantly reducing the probability of misidentifying oil, water



(polymer) as rock particles. The global accuracy can reach 0.99, and the IoU of the three classes is above 0.98, showing strong generalization ability and robustness.

2. This study is based on the idea of image processing, combined with pore distance transformation and watershed algorithm to complete the division of the control area of different radius pores in microscopic images, and proposes a method for characterizing the pore sweep efficiency, which can achieve quantitative and refined research on the recovery factor.
3. The results show that compared to water flooding, polymer flooding can increase the recovery factor by 24.01%, in microscopic experiments. Polymer flooding significantly improves the sweep efficiency of all levels of pores and throats, improving the uniform sweep of the displacement phase and making the utilization of pores more uniform. The lower limit of swept pore changes from 25  $\mu\text{m}$  for water flooding to 16  $\mu\text{m}$  for polymer flooding.

## Acknowledgements

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## Nomenclature

|             |                                           |
|-------------|-------------------------------------------|
| <i>SGD</i>  | Stochastic Gradient Descent               |
| <i>SGDM</i> | Stochastic Gradient Descent with Momentum |
| <i>CA</i>   | Class accuracy                            |
| <i>MCA</i>  | Mean Class Accuracy                       |
| <i>BS</i>   | BF(Boundary F1) Score                     |
| <i>MBS</i>  | Mean BF Score                             |
| <i>GA</i>   | Global Accuracy                           |
| <i>IoU</i>  | Intersection over Union                   |
| <i>MIoU</i> | Mean Intersection over Union              |
| <i>TP</i>   | True Positive                             |
| <i>FP</i>   | False Positive                            |
| <i>TN</i>   | True Negative                             |
| <i>FN</i>   | False Negative                            |
| <i>RF</i>   | Recovery Factor                           |

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